

Geovision: Country-Level Image Geolocation with Vision Transformers

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Introduction

Pinpointing where a single street-view frame was shot is hard even for seasoned *GeoGuessr* players, yet doing so automatically would unleash fast fact-checking and crisis-mapping tools. Humans subconsciously fuse many signals; an effective model must do the same.

Visual cues humans exploit

- **Climate & vegetation:** palms vs. pines, savanna vs. tundra
- **Architecture:** tile roofs, stilt houses, glass towers
- **Language on signs:** Latin, Cyrillic, Amharic, Hangul
- **Road culture:** paint colour, driving side, curb stones

The game's frequent mistakes—e.g. confusing *southern Brazil* with *northern Portugal*—motivate our **country-level** formulation. Given an RGB image $I \in \mathbb{R}^{H \times W \times 3}$ we predict a label $y \in \{1, \dots, K\}$.

Benchmarks we compare include humans (5 players), ResNet-50 CNN baseline, GPT-4o vision (zero-shot ceiling), and ViT-B/16(ours).

Country granularity balances *speed* and *actionability*: precise enough for regional analysis yet light enough for on-device inference. Our goal is to show that a compact ViT can beat humans and CNNs while narrowing the gap to GPT-4o, advancing reliable automated geolocation.

Background: Related Work

Early systems such as *Im2GPS* [?] used hand-crafted color/texture features plus nearest neighbors. The CNN watershed *PlaNet* [?] reframed geolocation as a 26 k-class problem, achieving city-level accuracy but at high compute cost and with coverage bias.

Country-level prediction strikes a clean trade-off: **VIPPGeo** showed a ResNet can directly predict country labels, while **PIGEON** introduced interpretable “semantic geocells.” Vision Transformers further improve context aggregation (TransLocator) and, when paired with language pre-training, enable strong zero-shot localization (GeoCLIP, StreetCLIP).

Milestones

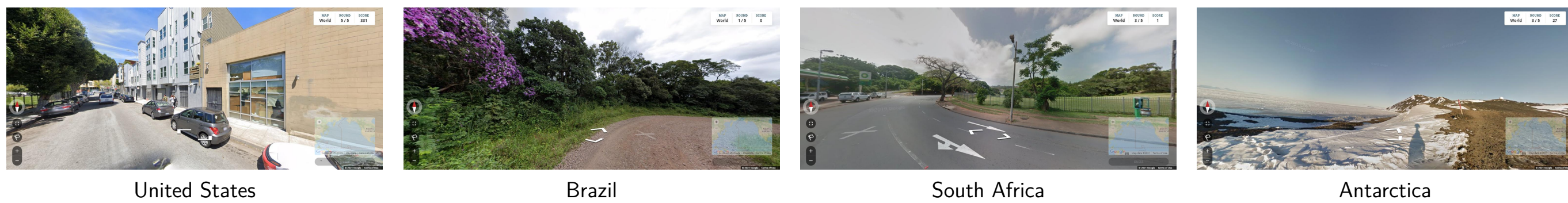
- **Im2GPS:** hand-crafted + NN
- **PlaNet:** 26 k geocells w/ CNNs
- **Transformers:** TransLocator
- **Country-CNNs:** VIPPGEO, PIGEON
- **Multimodal:** GeoCLIP, StreetCLIP, G³

Takeaway. CNNs raised the bar; **ViTs + lightweight multimodal cues now define the frontier**, motivating our ResNet-50 vs. ViT-B/16 study.

Dataset

We use the GeoGuessr image dataset from Kaggle [?], consisting of 49,997 RGB screenshots from the game. Each 1536×662 image is labeled with its country of origin (124 total), though distribution is highly imbalanced (median = 81 per country; max = 12,014). We perform an 80/10/10 stratified train/val/test split, excluding countries with only one image from val/test. Minimal preprocessing was applied—no cropping, augmentation, or normalization—to preserve spatial cues such as sign orientation and sun position.

The dataset reflects real GeoGuessr gameplay: images include visual noise like interface elements (score-card, compass, blurred minimap) and wide variation in scene content. Some images offer rich context (urban signage, vegetation, architecture), while others depict ambiguous roads, fields, or skies. This diversity makes the task more realistic—and more difficult.



Technical Approach

Problem Formulation We cast GeoGuessr image localization as a **104-way classification** task. Each RGB image $I \in \mathbb{R}^{H \times W \times 3}$ is mapped to logits $f_{\theta}(I) \in \mathbb{R}^{104}$. Models are trained to minimize cross-entropy loss:

$$\mathcal{L}_{\text{CE}} = - \sum_{k=1}^{104} \mathbb{I}[y = k] \cdot \log \left(\frac{\exp(f_{\theta}(I)_k)}{\sum_j \exp(f_{\theta}(I)_j)} \right)$$

We also experimented with Focal Loss to mitigate class imbalance, but standard cross-entropy led to better accuracy.

Human Baseline We collected predictions from 5 experienced GeoGuessr players across 100 held-out images. Participants gave both top-1 and top-5 guesses. This baseline reflects intuitive geographic reasoning using cues like road markings, vegetation, and signage—without data-driven consistency.

LLM Baseline: GPT-4o (Zero-Shot) We queried GPT-4o Vision with our test images using prompts designed to elicit structured JSON output containing top-1 and top-5 predicted countries. While the model received no fine-tuning for geolocation, its vision–language pretraining yielded strong results, serving as an upper bound.

Model 1: Finetuned ResNet-50 Architecture: We use microsoft/resnet-50 v1.5 from Hugging Face, initialized with ImageNet weights. The 1000-way classifier head is replaced with a new linear layer for 104 countries (random init). *Training Setup:*

- All convolutional layers are finetuned (not frozen)
- AdamW optimizer, LR = 5×10^{-5} , batch size = 32, 5 epochs
- Mixed-precision (FP16) enabled
- Stratified 80/10/10 train/val/test split

Processing: Images are resized, normalized, and tensorized via Hugging Face's AutoImageProcessor. *Remarks:* Despite strong generalization from ImageNet features, the model struggles with visually ambiguous or underrepresented regions.

Model 2: Finetuned Vision Transformer (ViT-B/16) Architecture: We use google/vit-base-patch16-224, pretrained on ImageNet-21k. Images are split into $14 \times 14 = 196$ non-overlapping 16×16 patches, projected into token embeddings. A [CLS] token aggregates global context across transformer layers. *Adaptation:*

- Replace classification head with new 104-class linear layer
- Full-model finetuning (not just classifier head)

Training Setup:

- 10 epochs, AdamW, LR = 2×10^{-5} , batch size = 32
- Hyperparameters tuned via short 2-epoch sweeps (LR, weight decay)
- FP16 training and weight decay = 0.01

Remarks: ViT lacks inductive biases (e.g., translation equivariance) but compensates with a large global receptive field. This enables long-range pattern capture (e.g., skyline–vegetation–road co-occurrence). End-to-end tuning allowed ViT to adapt to GeoGuessr’s noisy UI overlays and scene variability more effectively than CNNs.

Implementation All training used Hugging Face Transformers and Datasets libraries. Training pipelines used Trainer APIs with stratified splits and consistent preprocessing between models.

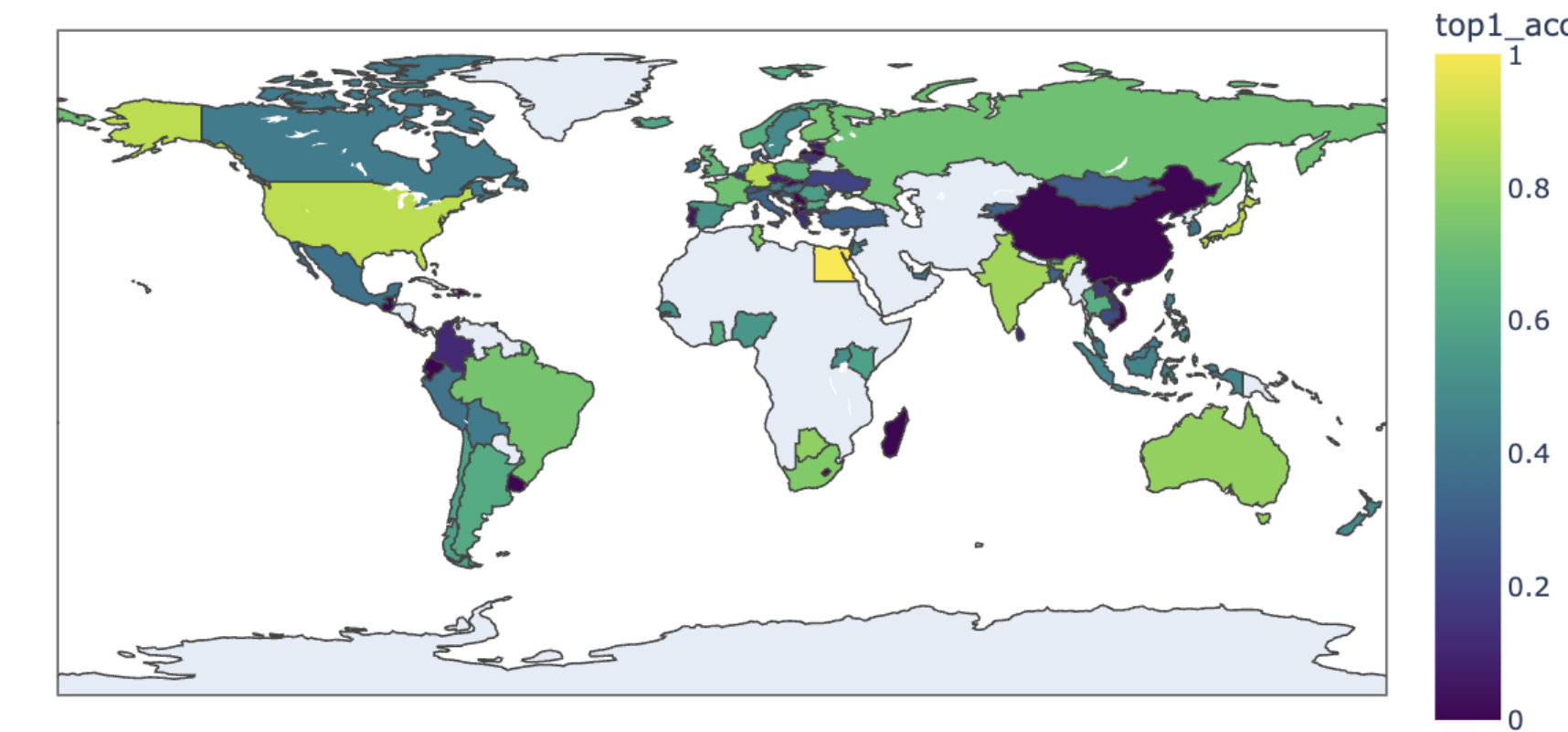
We evaluate using top-1 and top-5 accuracy:

$$\text{Top-1} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[\hat{y}_i = y_i], \quad \text{Top-5} = \frac{1}{N} \sum_{i=1}^N \mathbb{I}[y_i \in \hat{\mathcal{Y}}_i^{(5)}]$$

Findings Summary

Model	Top-1	Top-5
Human Baseline (n=5)	13.2%	31.6%
GPT-4o Vision (0-shot)	71.5%	86.5%
ResNet-50 (ours)	41.3%	59.2%
ViT-B/16 (ours)	67.5%	88.3%

- **ViT outperforms ResNet** by **+26 pp (Top-1)** and **+29 pp (Top-5)**.
- **ViT even surpasses GPT-4o** on Top-5 accuracy despite having <0.1% of its parameters.
- Human accuracy reflects task difficulty; all models outperform by $3 \times - 5 \times$.



ViT-B/16 Top-1 accuracy by country

Model-Specific Analysis

- **GPT-4o** excels in digitally visible regions (e.g. US, India, Europe) thanks to web-scale training.
- **ViT** performs well in regions with visual cues like architecture, soil, vegetation, or signage.
- **ResNet** struggles in ambiguous regions (Portugal, Spain) due to reliance on local textures.
- **Low-resource countries** with <50 images (e.g. Laos, Lithuania) see top-1 accuracy < 20%.
- **No-shot classes** (e.g. Uruguay, Slovenia) are predictably misclassified—showing need for class-aware balancing.



NYC example (ViT, ResNet): USPS + NYPD cues; all humans were correct. **Brazil frame (both):** Red soil + palm-lined road helped disambiguate.

Conclusion

Limitations. Our dataset underrepresents countries with limited Street View access, skewing exposure toward high-income regions. Country-level prediction masks regional ambiguity, such as Belgium vs. Netherlands, Brazil vs. Russia. Additionally, our human baseline comes from a small, homogeneous group (five U.S. students), limiting the generalizability of performance comparisons.

Conclusion. Transformer-based models – especially ViT-B/16—emerge as strong contenders for automated geolocation. Despite using far less data than GPT-4o, ViT outperformed it on **top-5 accuracy** (88.3% vs. 86.5%), indicating broader scene understanding. ViT’s advantage likely stems from its learned generalization, while GPT-4o benefits from web-scale memorization.

In applied settings like fact-checking or disaster response, a shortlist of likely countries is often more valuable than a single guess. Our results show that task-specific ViTs can offer this effectively and efficiently– closing the performance gap to frontier vision–language models with a fraction of the resources.